

The YAOKI Lunar Rover

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Webinar | Lunar Ledger | First Dataset Release

Launching
to space:

SpaceX
Falcon 9



PHOTO CREDIT: SPACEX

Leaving Earth:

Intuitive Machines
IM2 Mission



Leaving Earth:

Intuitive Machines
IM2 Mission



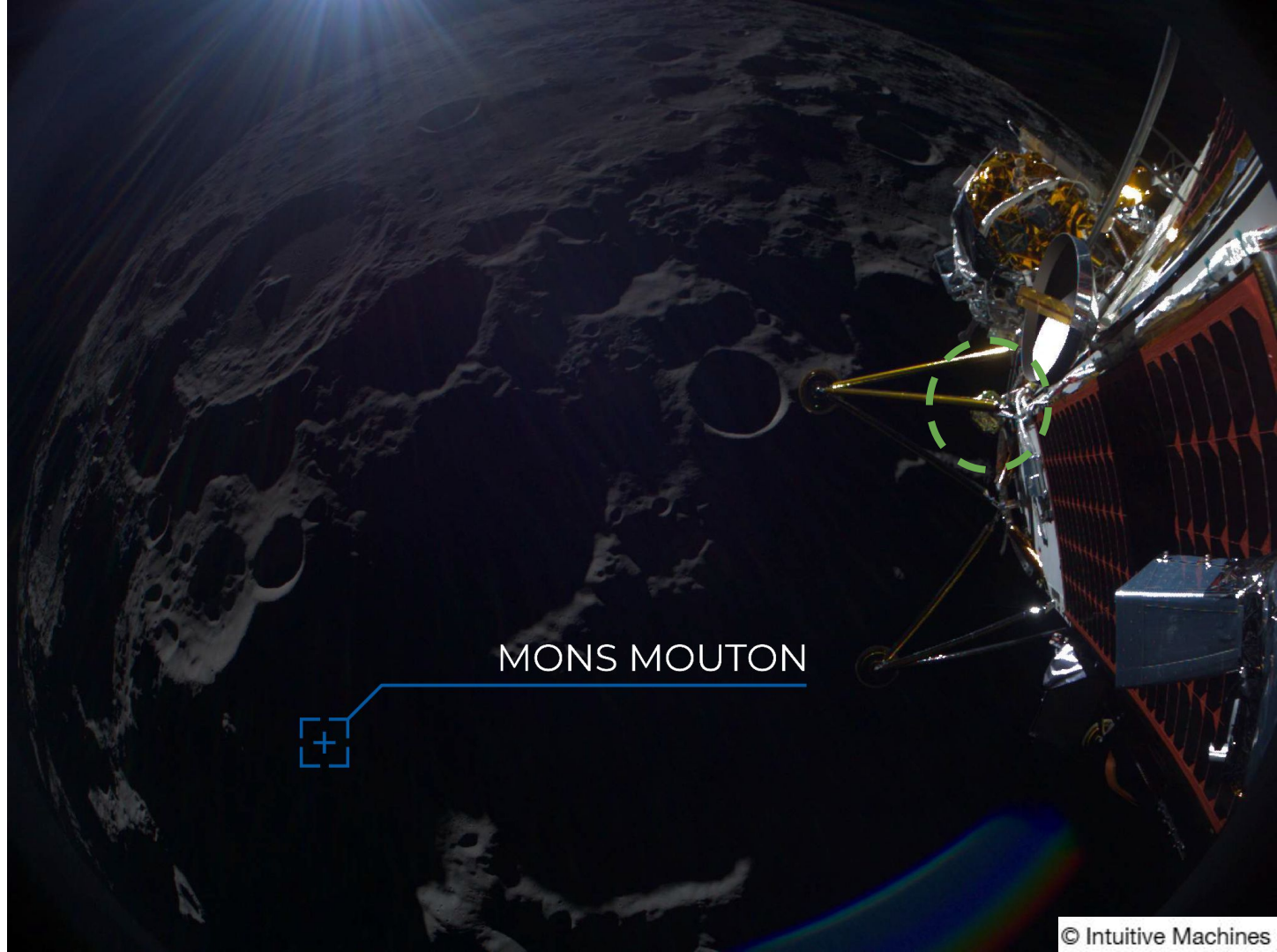
Arriving at the
Moon:

Intuitive Machines
IM2 Mission



Landing on the
Moon:

Intuitive Machines
IM2 Mission



Landing on the
Moon:

Intuitive Machines
IM2 Mission



LANDED
ON
THE
MOON
!!!



© Intuitive Machines

Landing Site

84.8 °S, 29.1°E

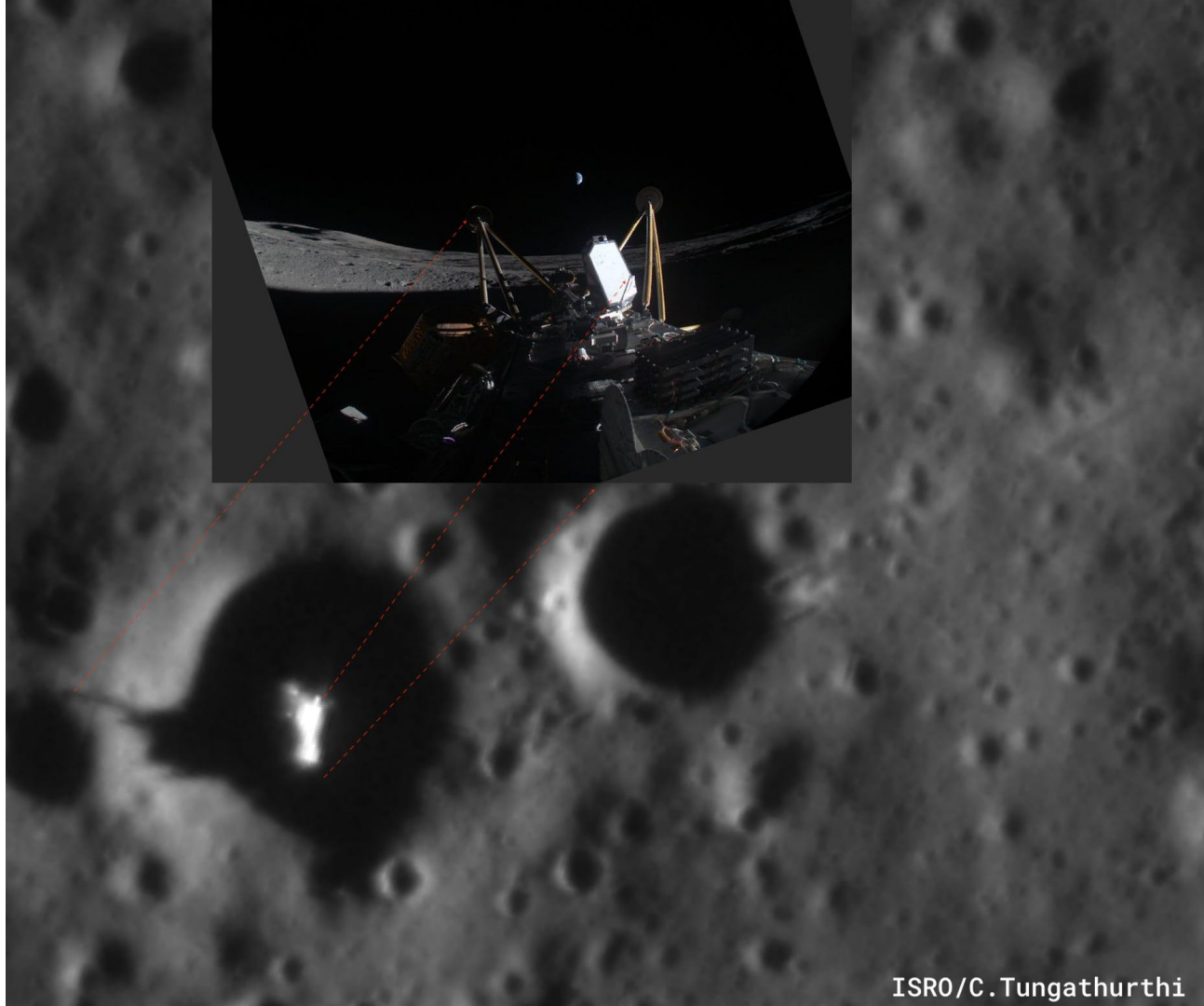
Distance from the
South Pole:

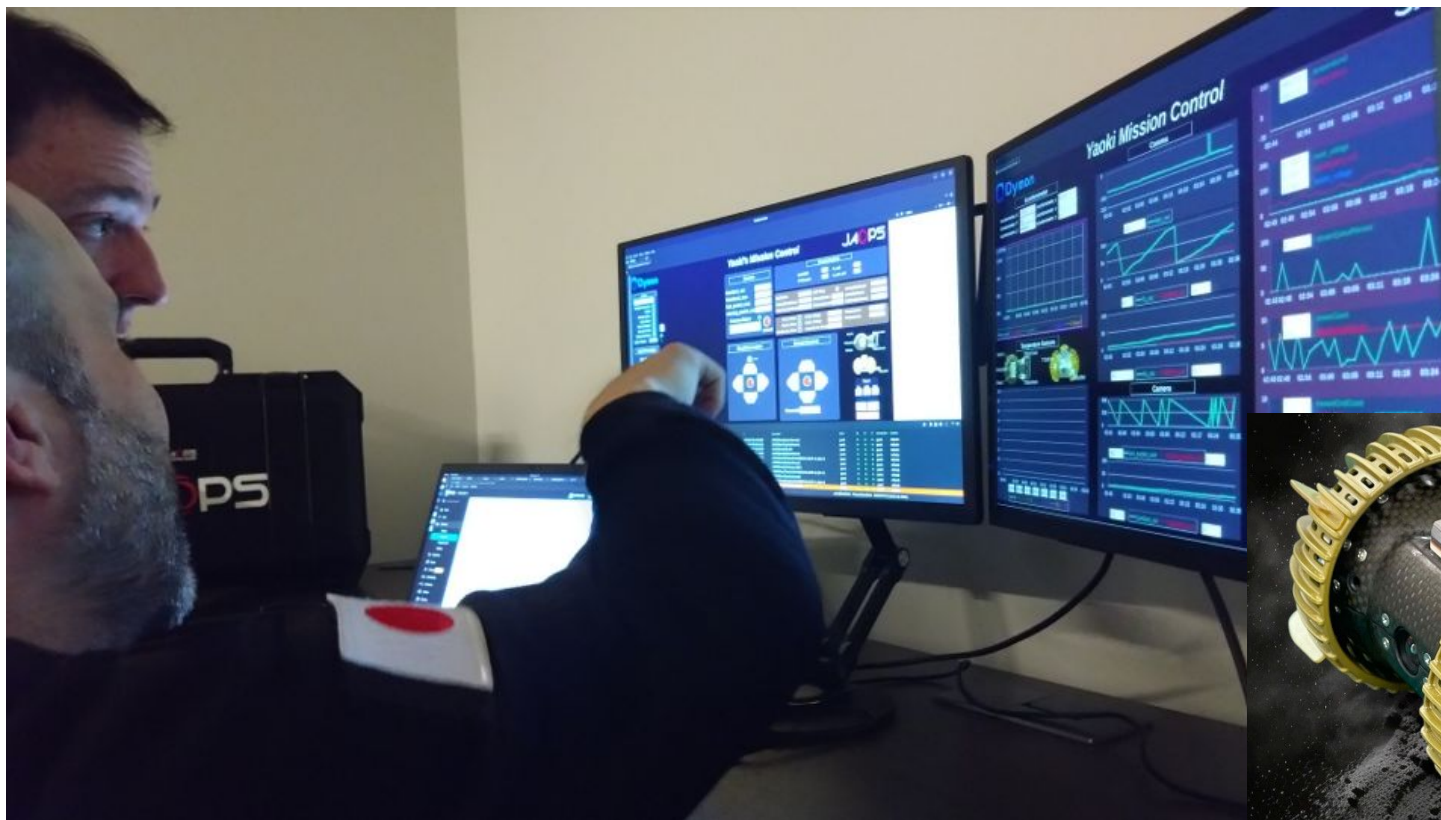
~160 km

Crater

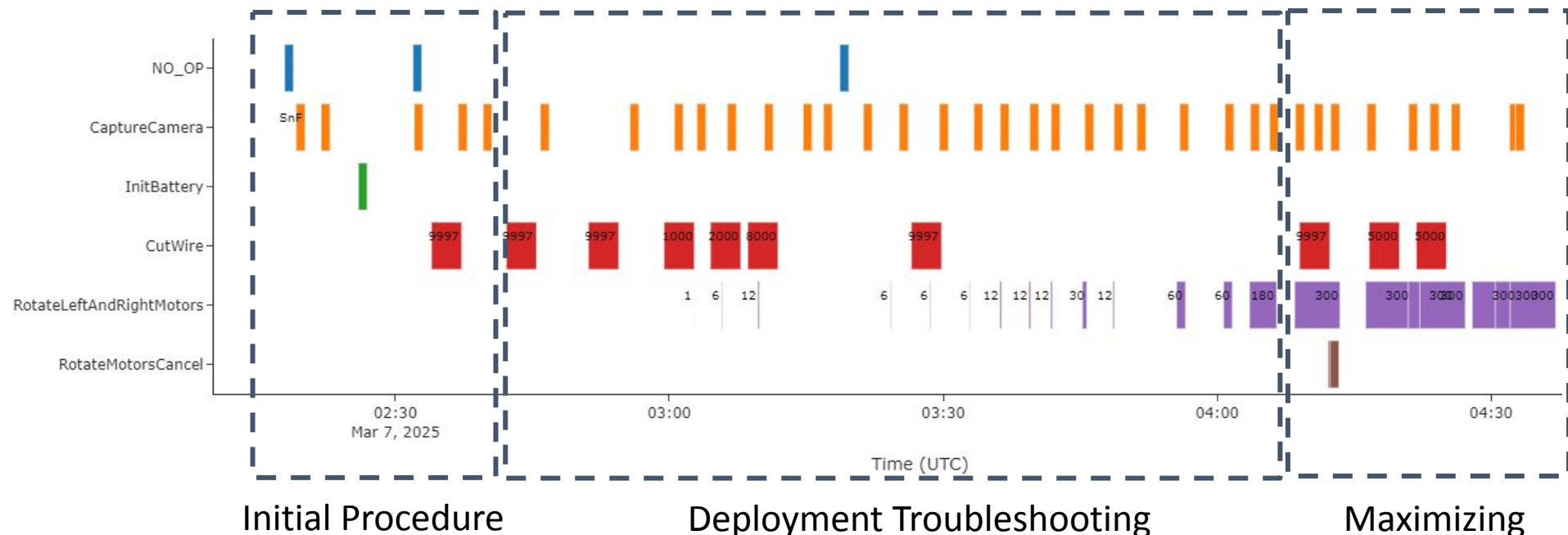
Temperature :

~ -100 °C

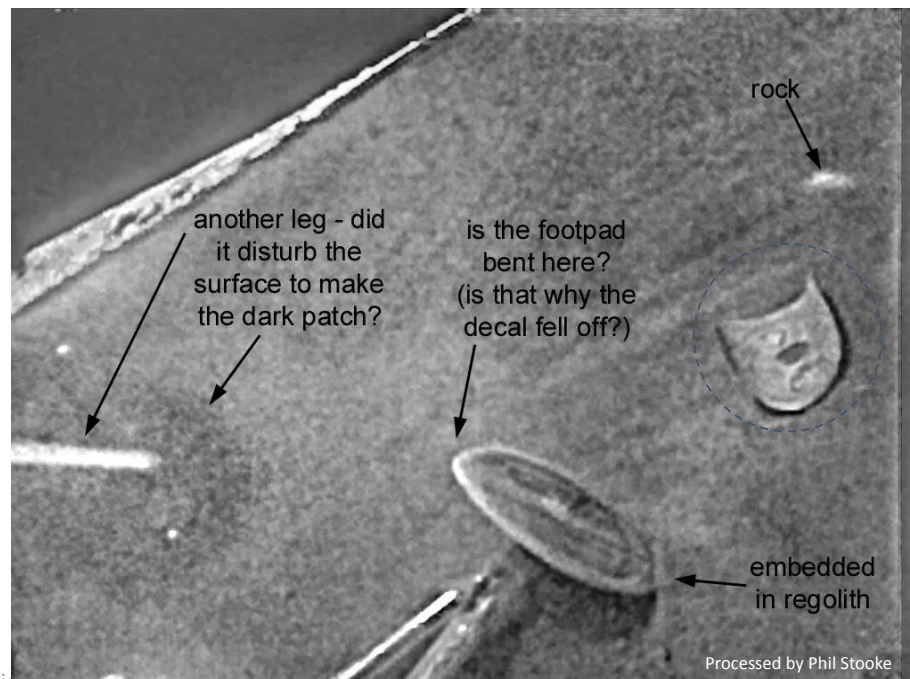




Timeline of Telecommands Sent



Images taken from the crater near the lunar South Pole



All 28 YAOKI Images:

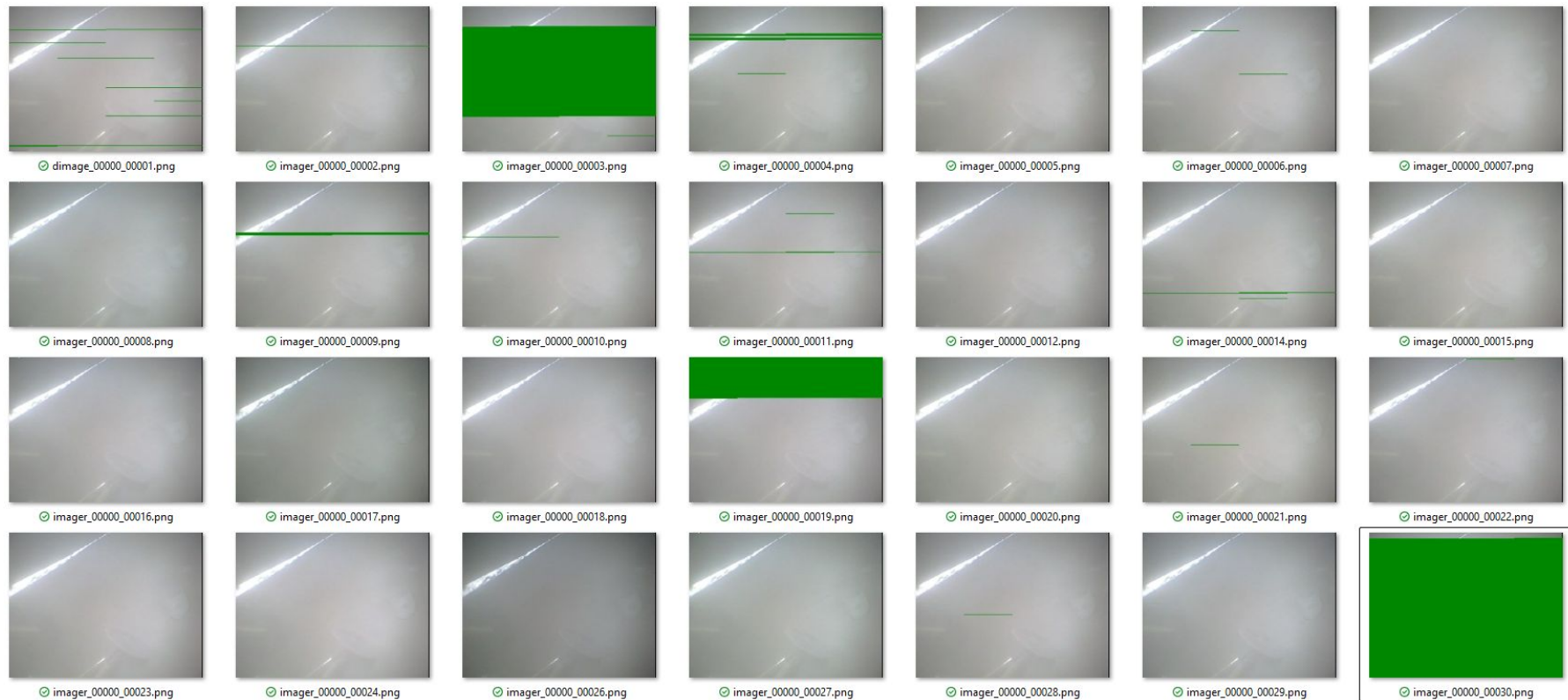
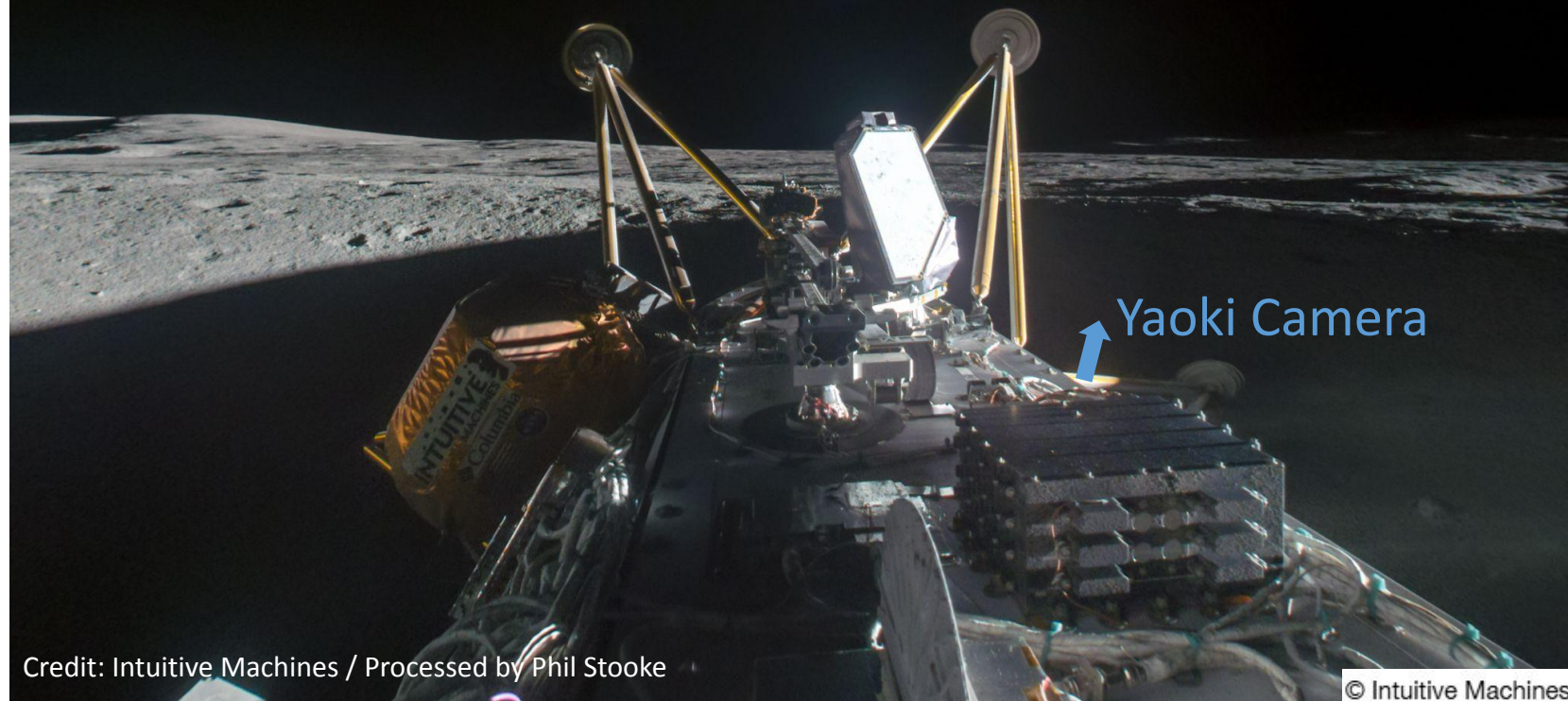


Image from IM2 lander camera

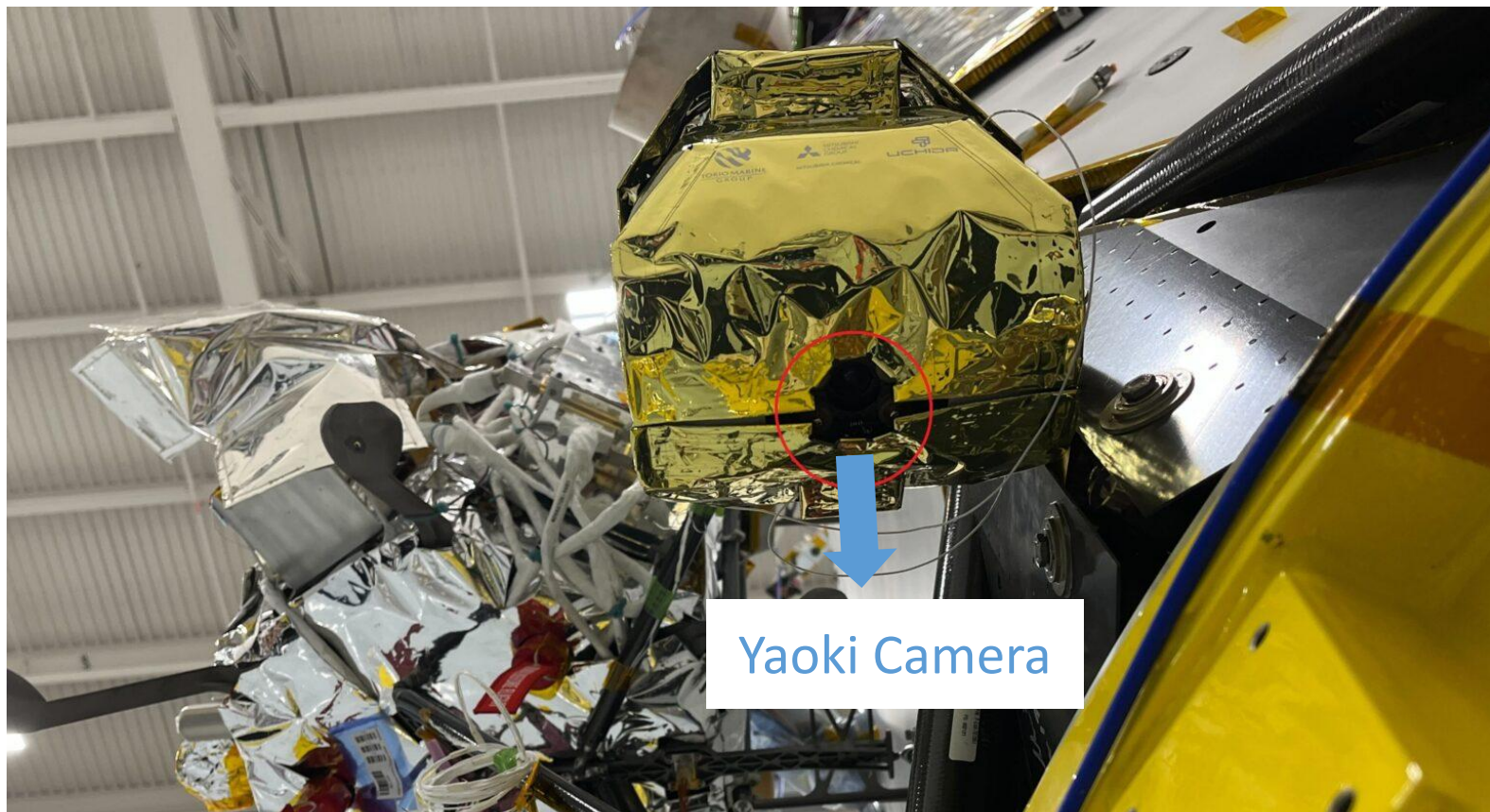


Credit: Intuitive Machines / Processed by Phil Stooke

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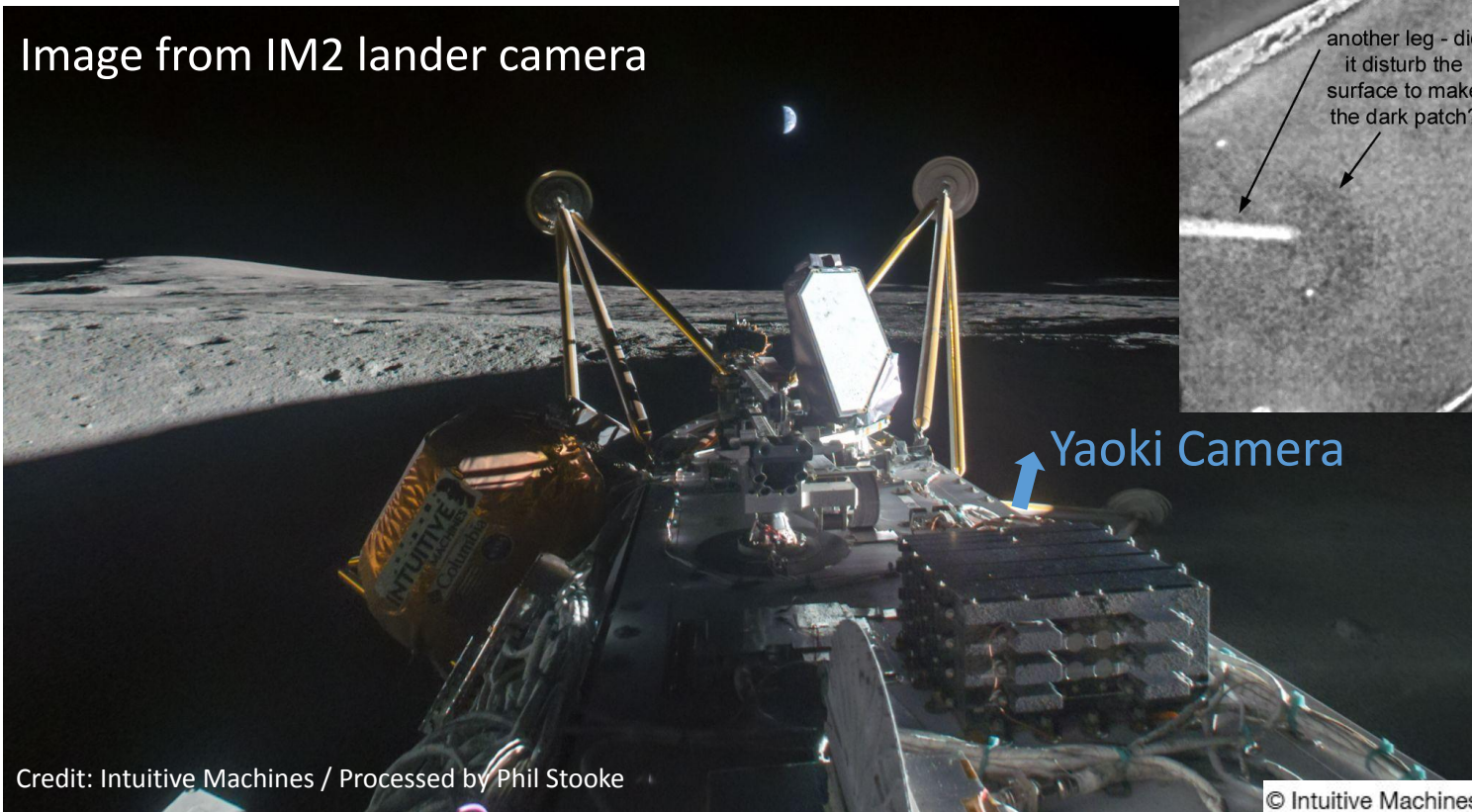
YAOKI
Rover
before
launch

Red Circle =
YAOKI
camera
view port



Yaoki Camera

Image from IM2 lander camera



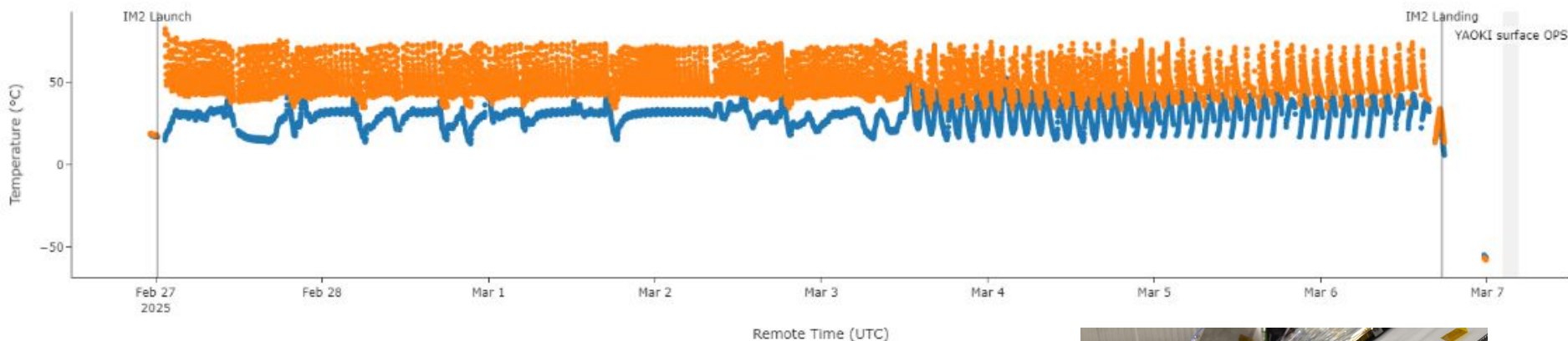
Credit: Intuitive Machines / Processed by Phil Stooke



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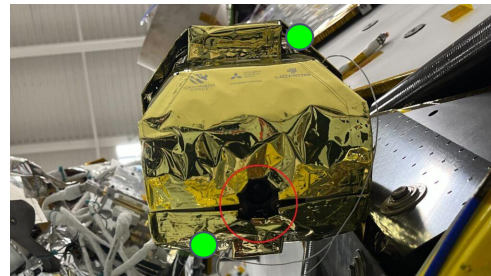
External Temperature Sensors (x2)

Temperature Telemetry over Time (cleaned)



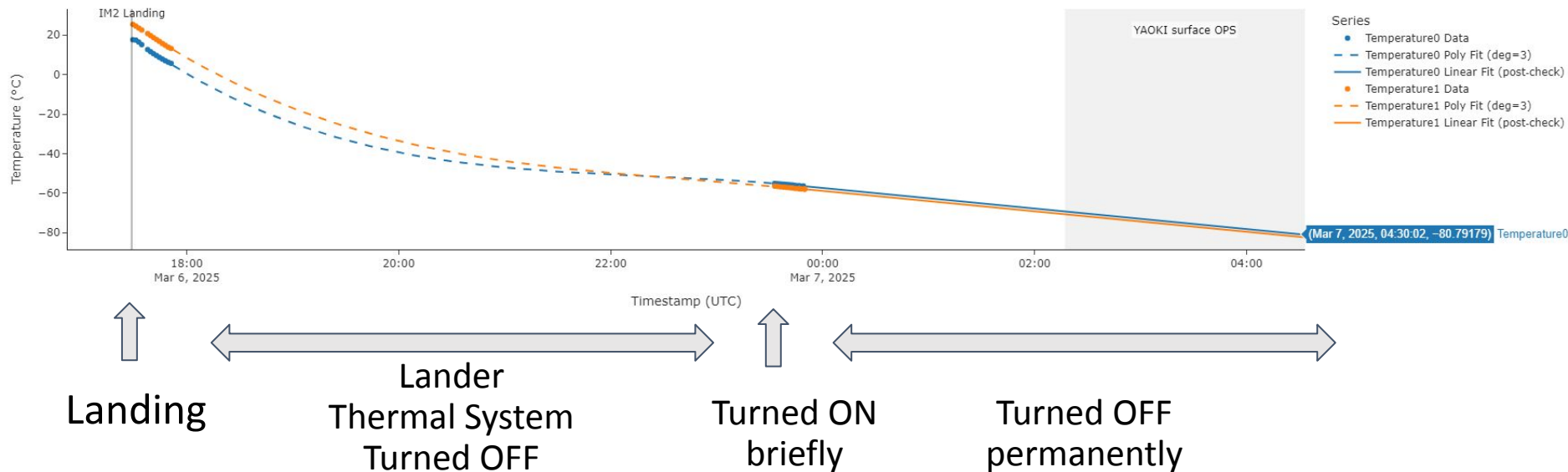
Green dots =

External temperature sensors

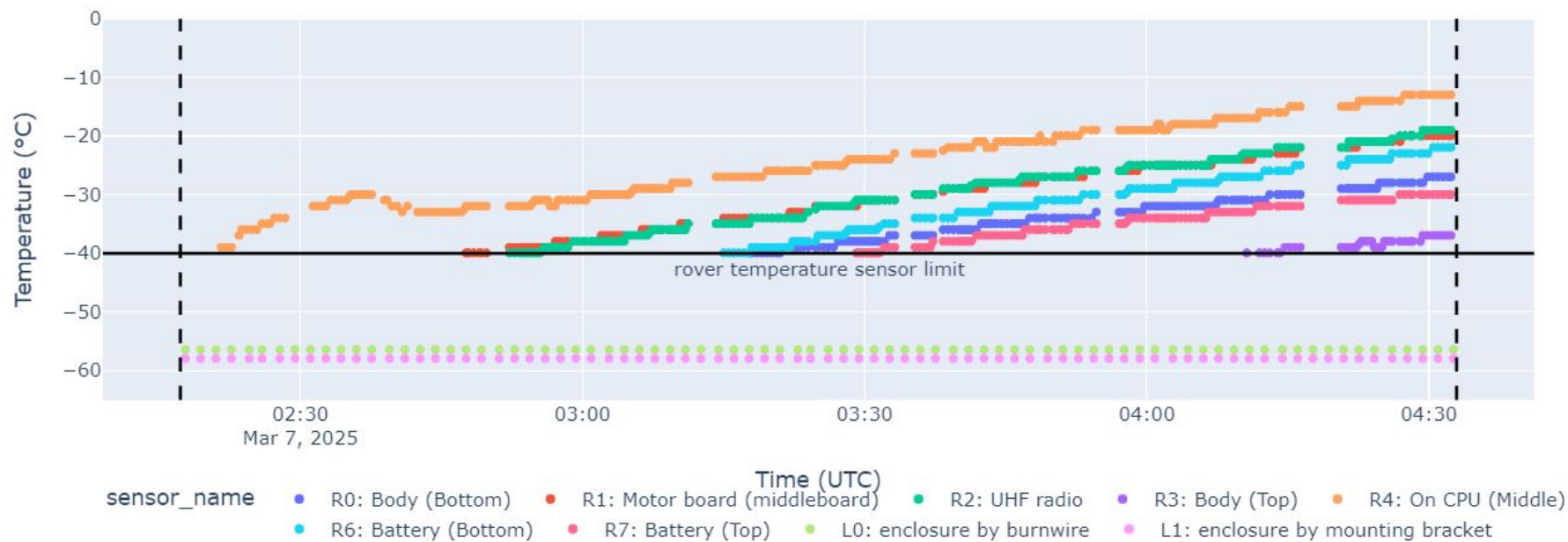


External Temperature Sensors (x2)

Temperature Fits: Polynomial (to last point) & Linear (post-check extrapolate)



Internal Temperature Sensors (x7)



- All telecommands sent
- All telemetry received
- All images received

Open Data on AWS



jaops-space / im2-yaoki-yamcs-public

- Raw format and processed format
 - Tools and tutorials of how to access and analyze the data
 - Description of each data item
-
- Configuration of the Yamcs Mission Control System
 - Mission Database
 - Data Archive
 - You can replay the data like it was received during the mission

Rover Name	Country (Entity)	Mission Name	Year
Lunokhod 1 and 2	Soviet Union 	Luna 17, 21	1970, 1973
Apollo Lunar Rovers	USA (NASA) 	Apollo 15, 16, 17	1971–72
Yutu and Yutu 2	China (CNSA) 	Chang'e 3, 4, 6	2013, 2019, 2024
Pragyan	India (ISRO) 	Chandrayaan-3	2023
SORA-Q (LEV-2)	Japan (JAXA) 	SLIM	2024
MAPP	USA (Lunar Outpost) 	Lunar Voyage 1 on IM-2	2025
YAOKI	Japan (Dymon) 	YAOKI on IM-2	2025

Only 10 successful rovers

Operated by JAOPS

First Japanese private-sector rover on the Moon
First images of dark crater near South Pole

**First lunar rover with open-source mission
operations software**

First dataset on the Lunar Ledger!

Surviving the launch, cruise and surface environments is hard

The accelerometer failed. But all other electronics worked well!

Not space-grade / multi-million dollar dedicated development

Sending data from the Moon is hard

Choose carefully what data to send, how much can be sent, how to send it

Always send raw data. Calibration only on the ground.

Software to re-configure data sending parameters

Landing on the Moon is hard

Record data in space before landing

Plan for contingency landing scenario together with the lander provider

Expect anomalies: design robust deployment system and data acquisition

Mission Operations are critical from the beginning of development

Hardware + Software + Mission Operations

Standards and heritage increase reliability and reduce development time

Yamcs Mission Control System

CCSDS communication protocols

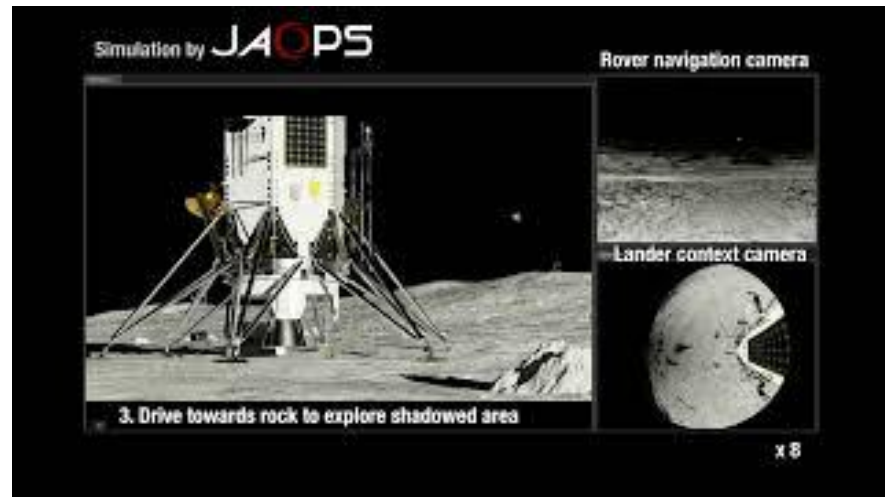
Deployment infrastructure

Test early, test often, test fully

Create (and budget) for several Flight Models

Leverage simulation from development to mission planning to testing to operator training to collecting scientific data properly

Expect and prepare for the unexpected





Why Join the Lunar Ledger?



1. Share this unique dataset
2. Share lessons learned
3. Empower others to analyze
4. Set a precedent

Vision of JAOPS:

JAOPS is committed to being a driver of safe, successful, and sustainable space mission operations by promoting best practices for the planning, collection, and delivery of lunar mission data.

Vision of Dymon:

Dymon continues to push the frontier through new missions that will collect valuable data and support humanity's expansion into space.

Vision of YAOKI:

Fall 7 times, stand up 8 times!